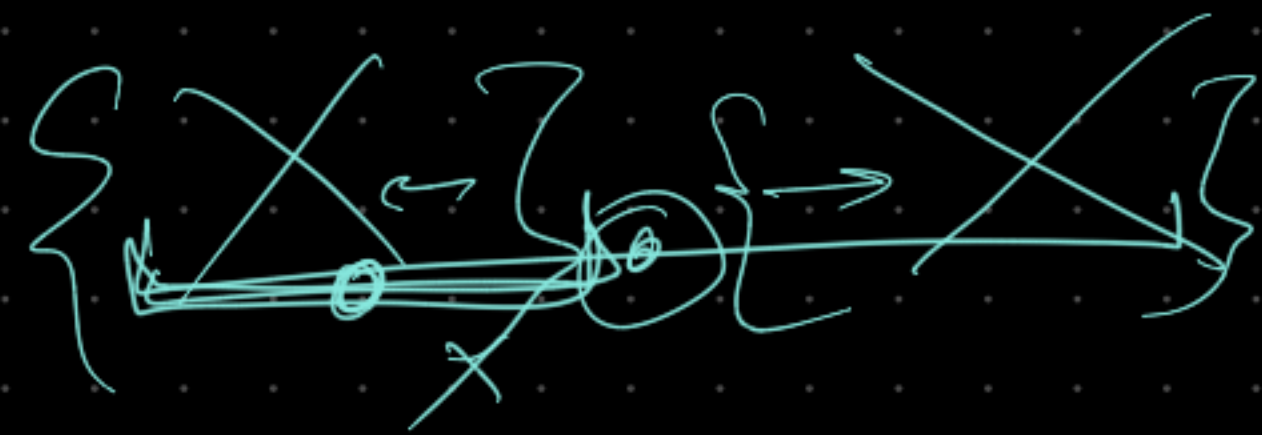


1000 words

unsorted $\rightarrow$ 1000 steps

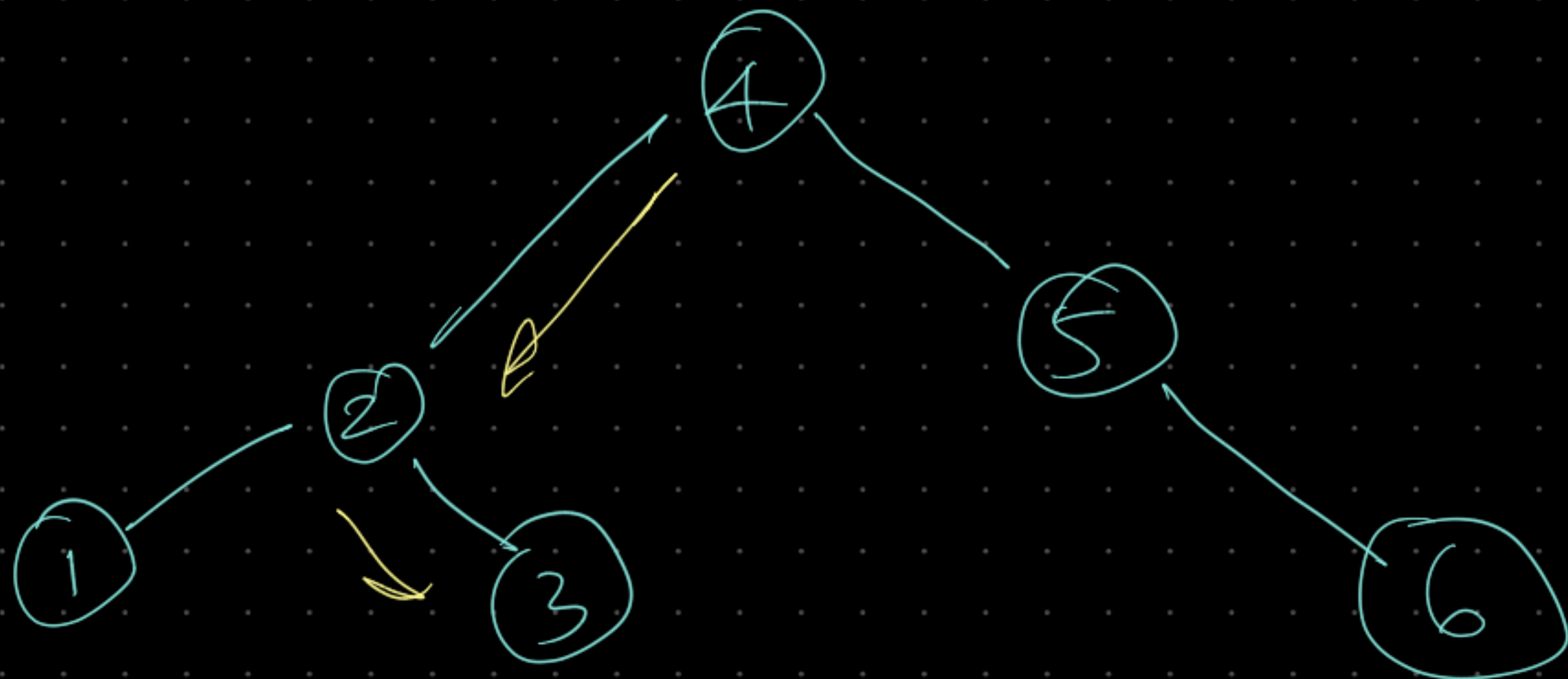
sorted $\rightarrow$ 10 steps

$\log_2(1000) = 2^x \rightarrow 1000$



$n \rightarrow \frac{n}{2} \rightarrow \frac{n}{4} \rightarrow \frac{n}{8} \dots$

1000 $\rightarrow$ 500 $\rightarrow$ 250 $\rightarrow$ 125 $\rightarrow$ 62
 $\rightarrow$ 31 $\rightarrow$ 15 $\rightarrow$ 7 $\rightarrow$ 3 $\rightarrow$ 1



Left sub-tree always small $<$ root

Right sub-tree always great $>$ root